



BY: YOUSEF ELSAYED, FIXED SOLUTIONS

Course Code : <https://github.com/yousef8/nodejs-course>

اليوم الأول

المحتويات

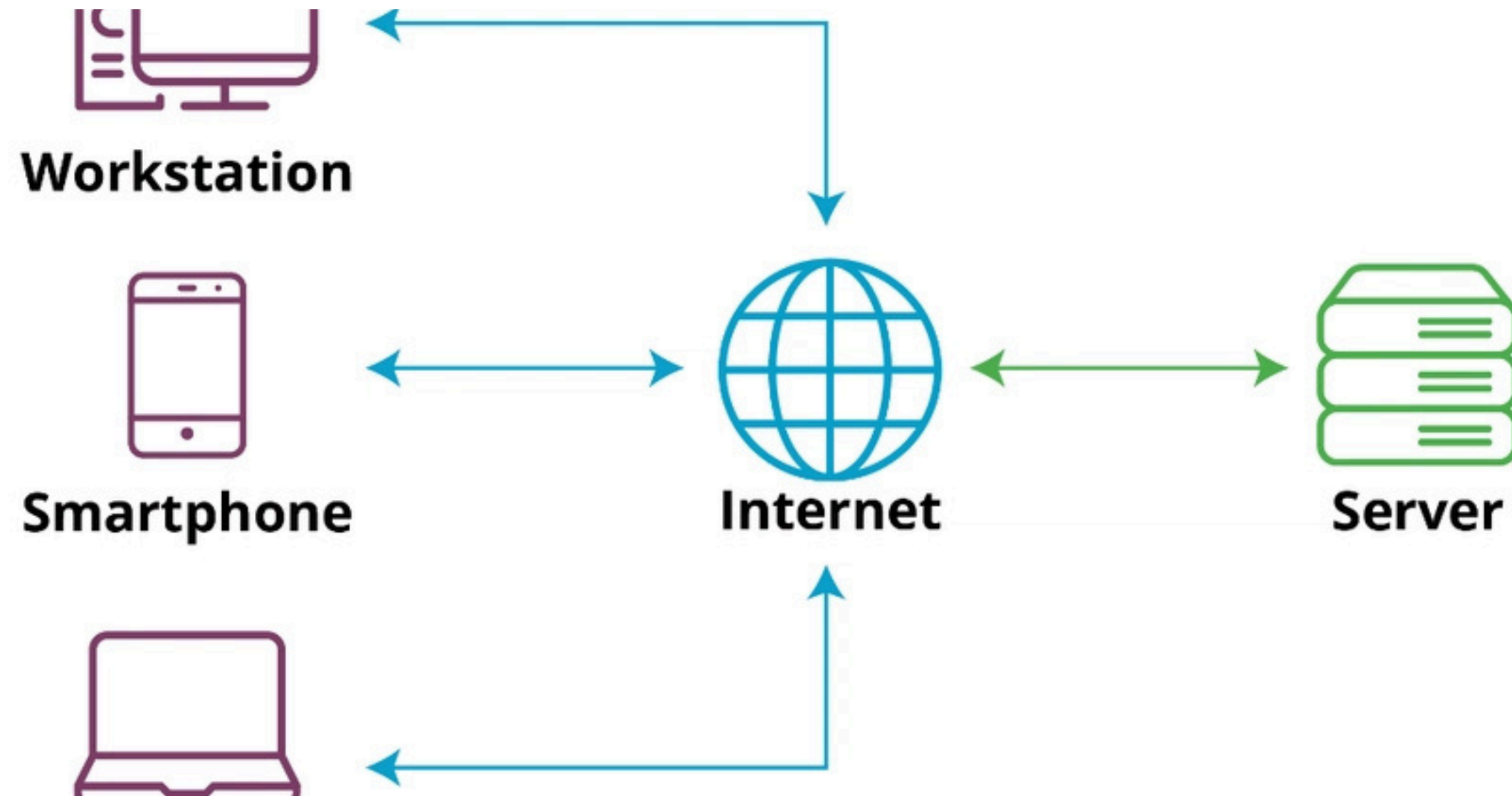
1. المتطلبات
2. مفاهيم أساسية
3. نشأة جافا سكريبت
4. نشأة nodejs
5. بيئة عمل nodejs
6. تثبيت nodejs
7. مدير الحزم npm
8. الوحدات (Modules)
9. مشروع قائمة المهام (todo)

المتطلبات

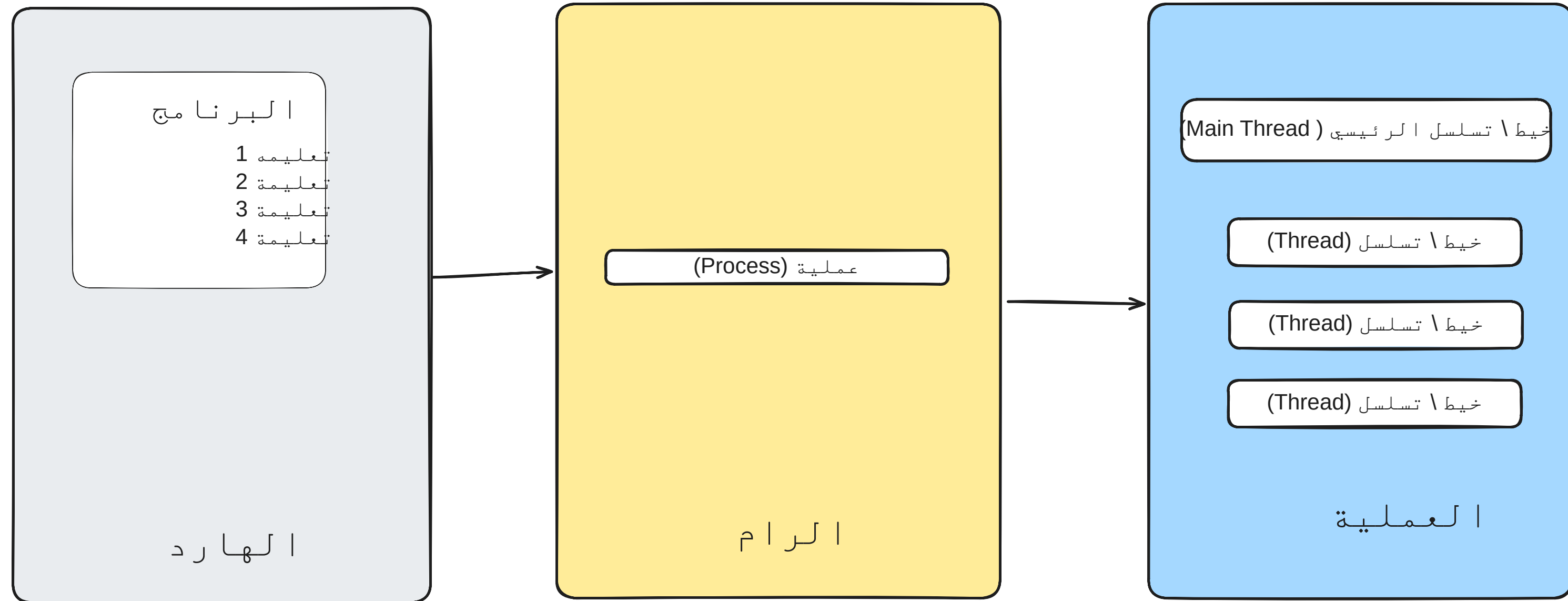
- جافا سكريبت
 - إيكماسكريبت 6
 - حلقة الحدث (event loop)
- لينكس

مفاهيم أساسية

Web 101



خيطة (thread)

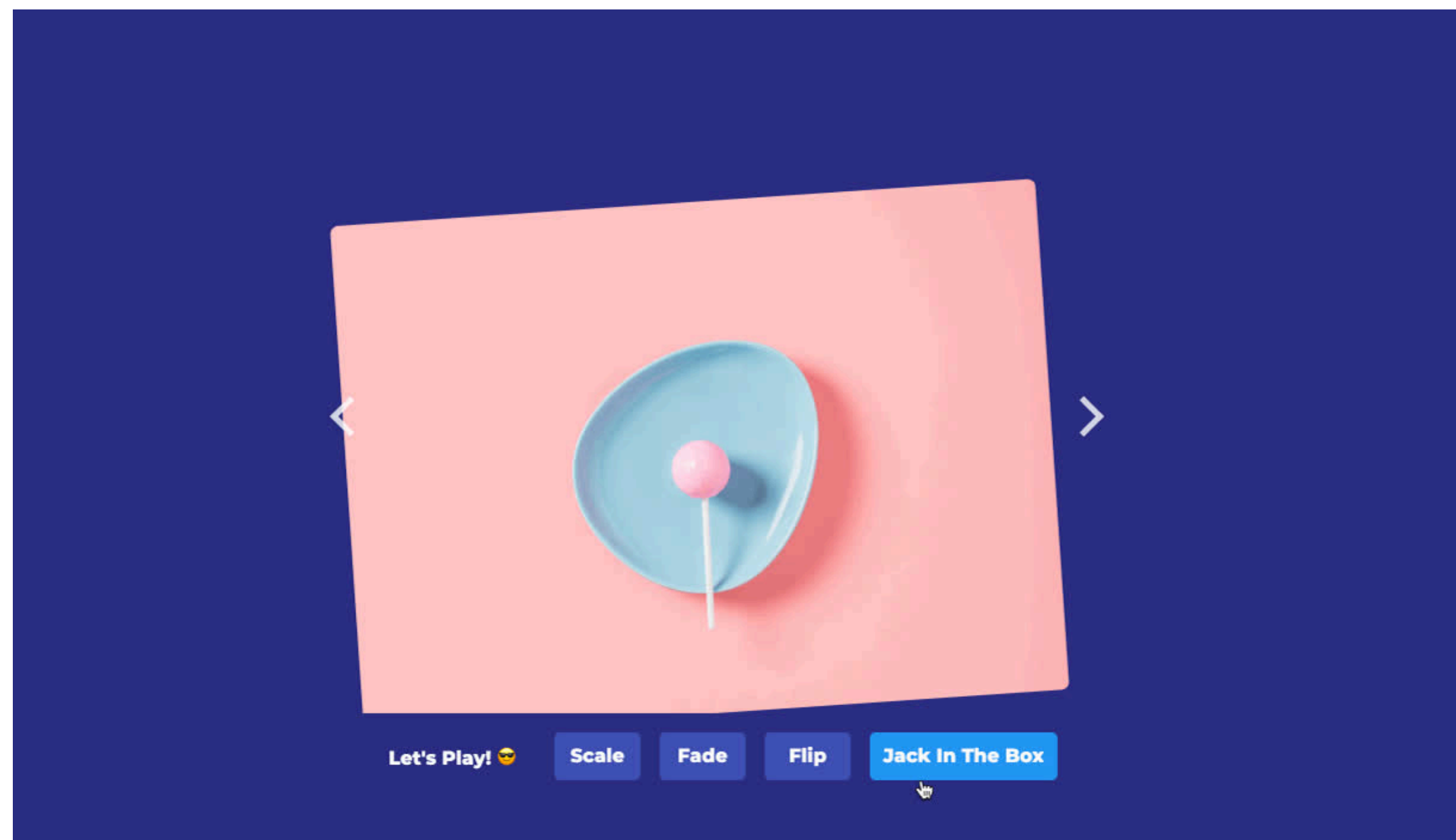


نشأة الجافاسكريبت

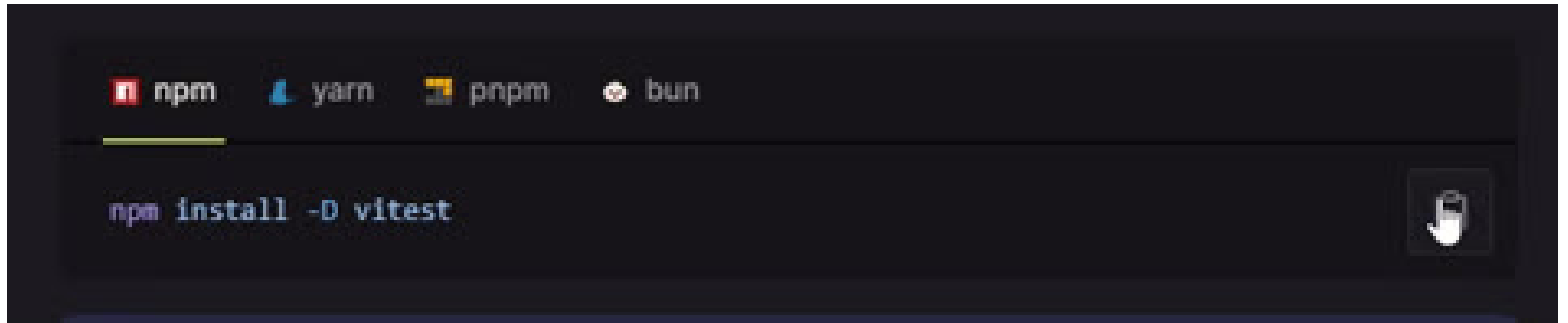


BRENDAN EICH

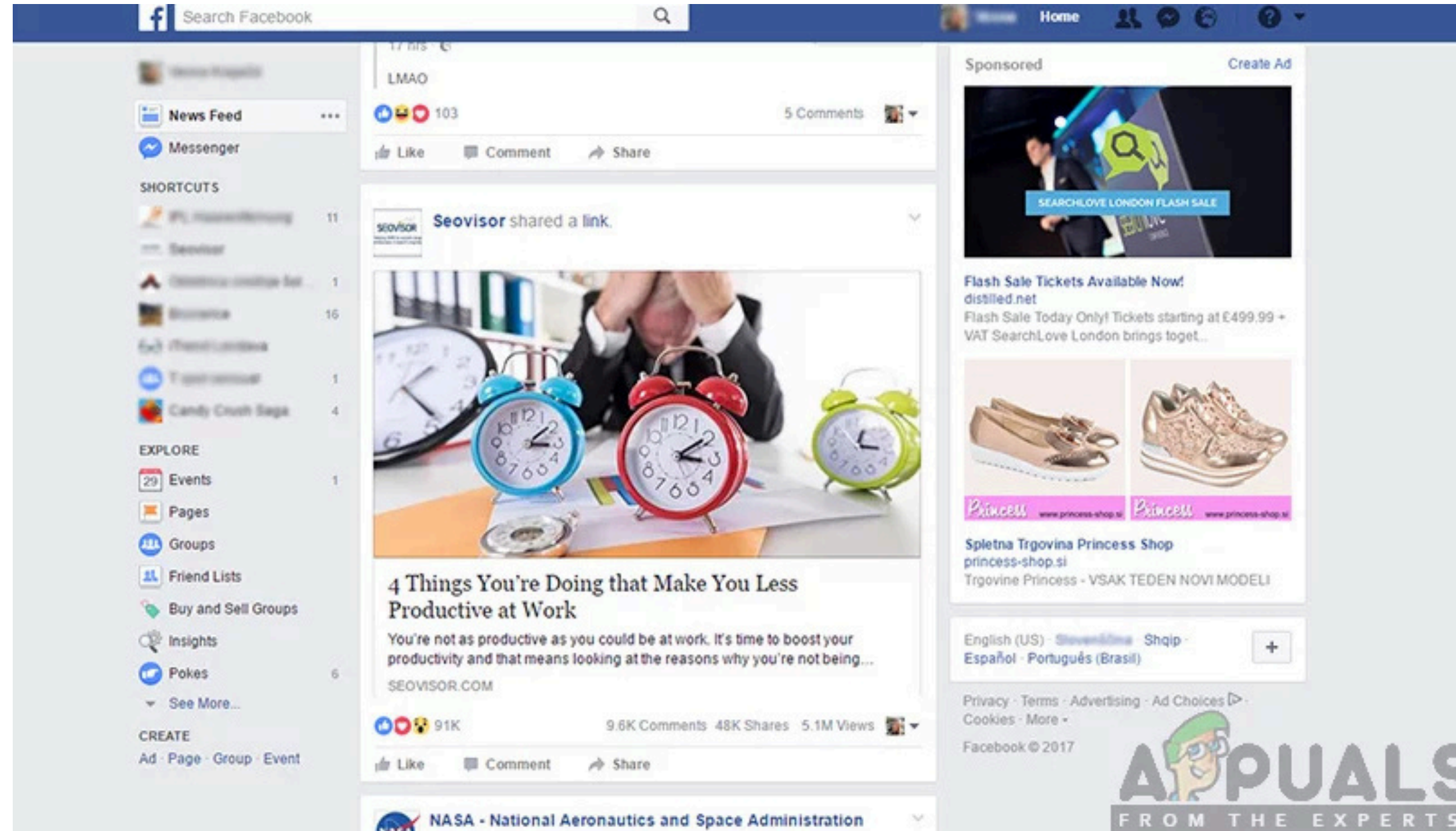




دوّارة الصور - IMAGE CAROUSEL



نسخ إلى الحافظة - COPY TO CLIPBOARD



التمرير اللانهائي - "INFINITE SCROLL"

ECMAScript



Chrome + Edge + Opera

 SpiderMonkey

FireFox

نشأة Node.js

All backend languages back then were synchronous

Every request meant a new thread

Didn't scale well for tens of thousands of concurrent connections



**JS , for it's popularity and being High level
v8, the new optimized engine from google
it doesn't have I/O module because it runs on browser**

**Node.js هو بيئة تشغيل (runtime) لجافاسكريبت
غير متزامنة (asynchronous)،
تعتمد على الأحداث (event-driven)،
وغير حاصبة (Non-blocking)،
مبنية على محرك V8 الخاص بمتصفح Chrome**

Node.js

JavaScript API

C++ wrapper

C libraries

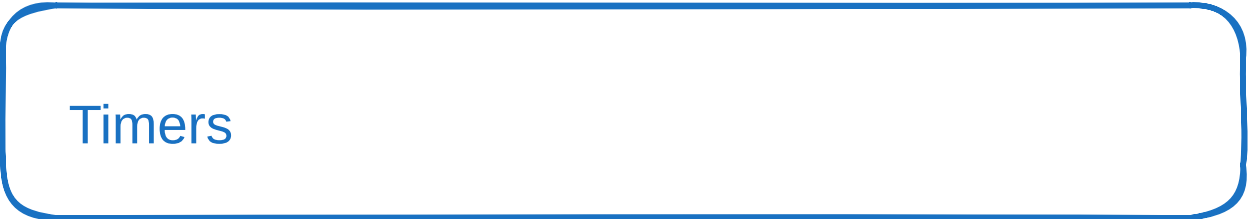
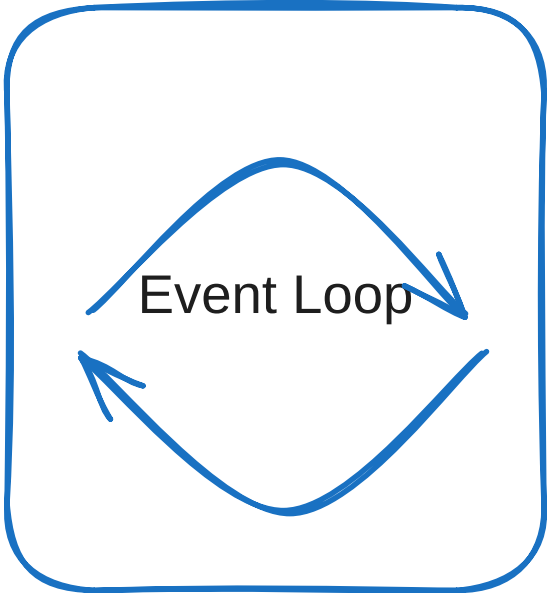
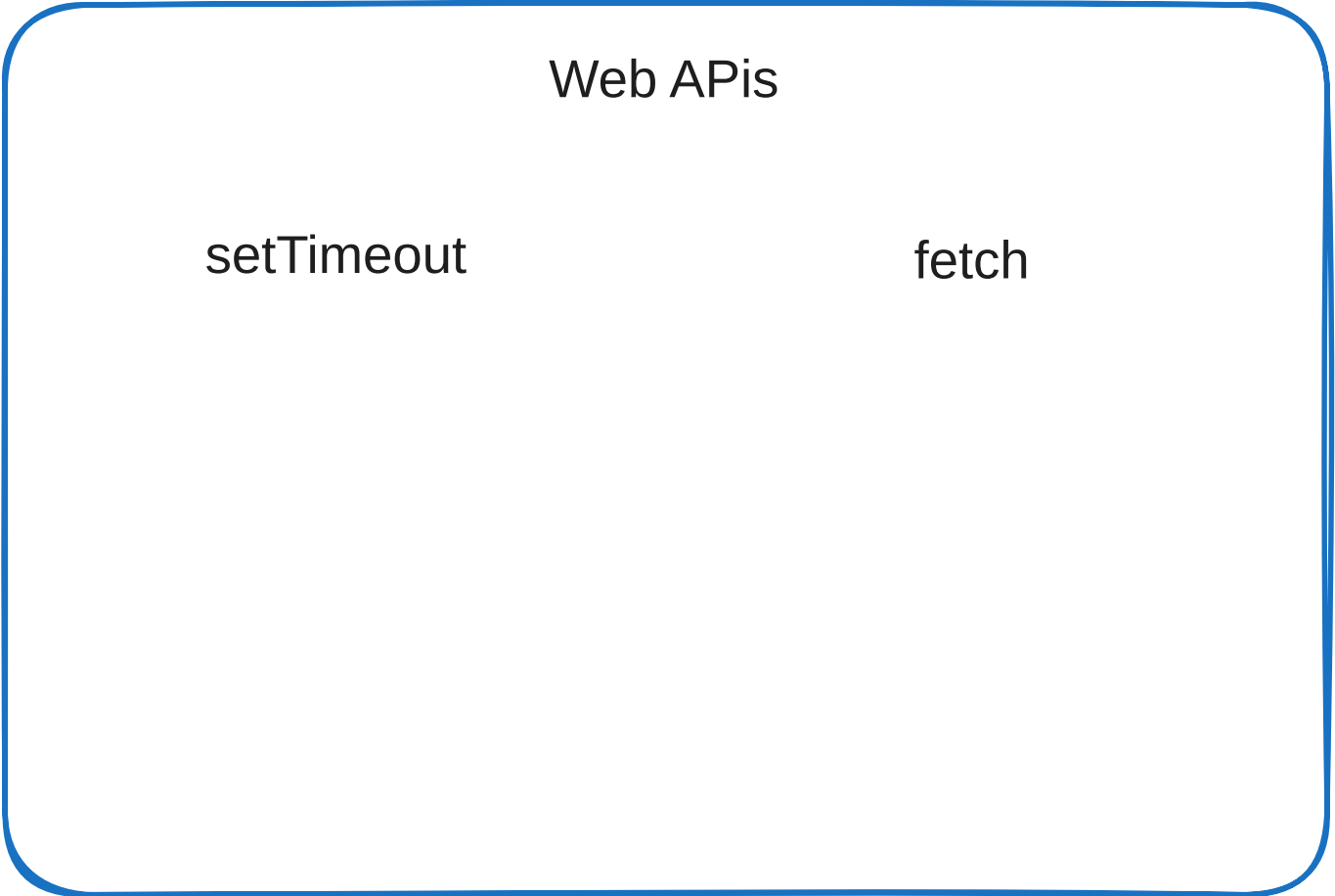
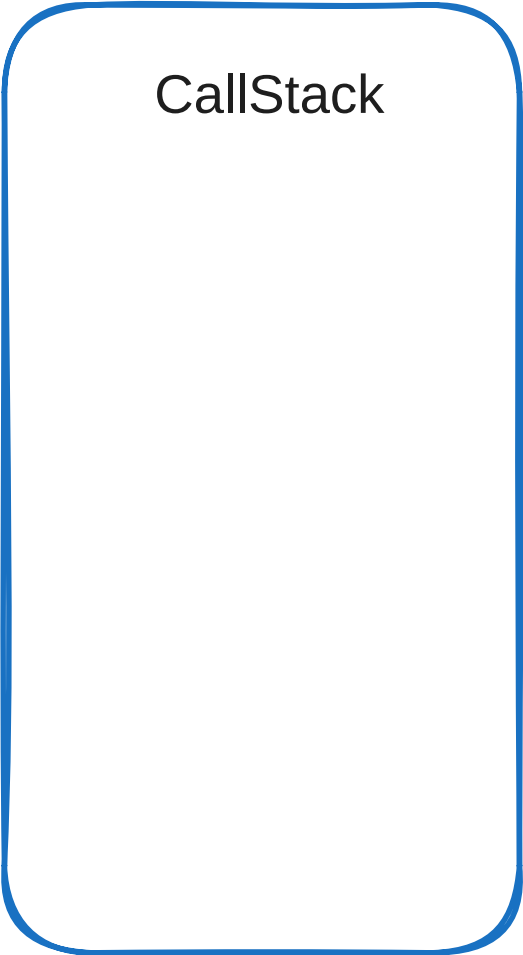


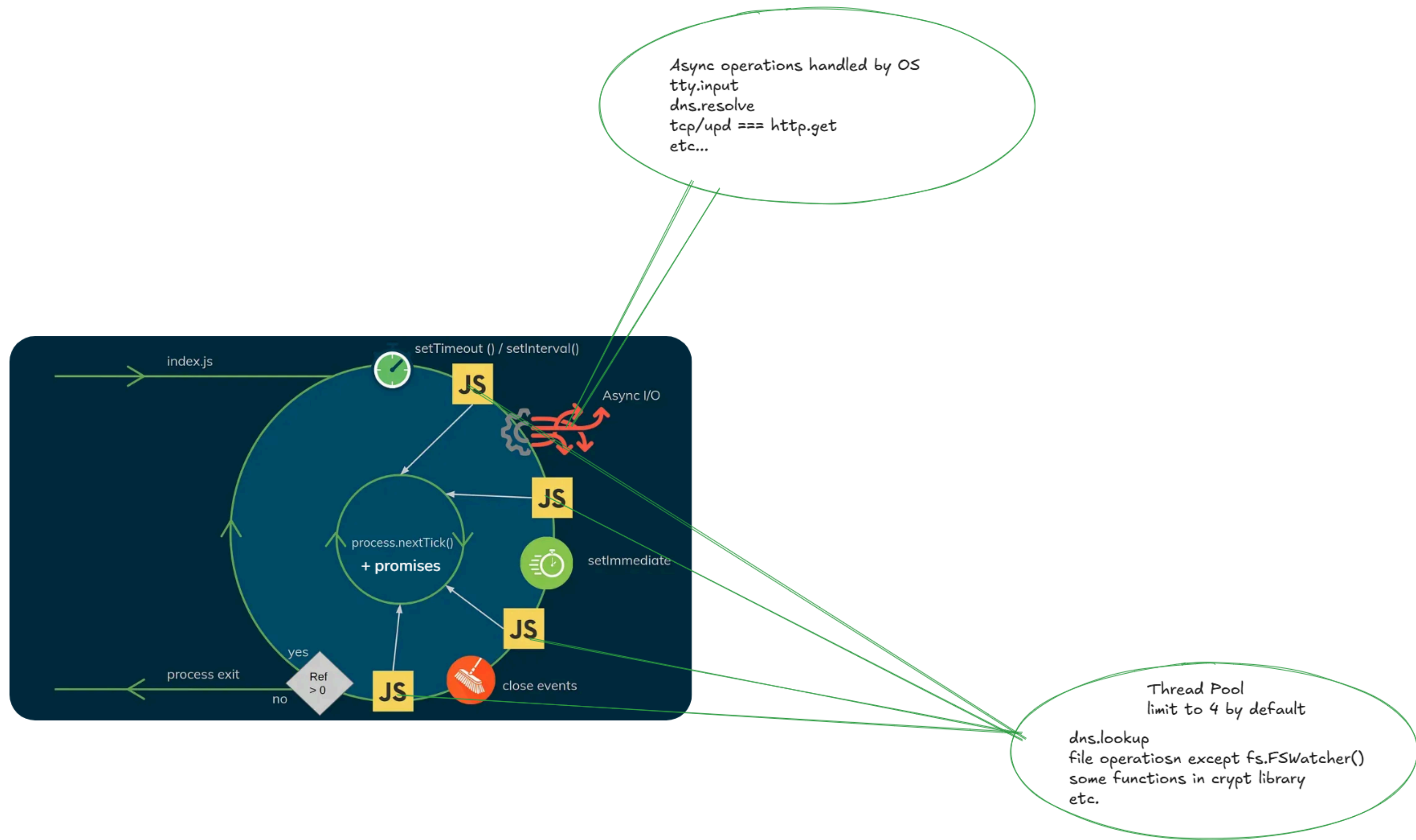
libuv
cross platform
asynchronous I/O



other libraries like:
OpenSSL or c-ares

Async code example





Browser.Js vs Node.Js

Node.js	Browser JS
	DOM
	Browser API
Accessing filesystem using <code>`fs`</code> module	
Creating servers using <code>`http`</code> module	
Interacting with DB	

Node.js use cases

- Best when you need **speed, scalability, and real-time** communication — especially for **I/O-bound** (input/output)
 - Real-time applications (chat apps, live dashboards, online games)
 - Streaming apps (video/audio streaming, file uploads)
 - Command-line tools / build scripts
 - Proxies / Gateways
- Node.js is **not great for CPU-heavy tasks** — anything that blocks the single main thread.
 - image/video processing, complex math, AI model training, encryption-heavy logic
 - Applications that need multithreading
 - Real-time 3D rendering/graphics apps (CAD applications)

تسطين

Release Cycle

2 . x . x

الإصدار الزوجي - المستقر

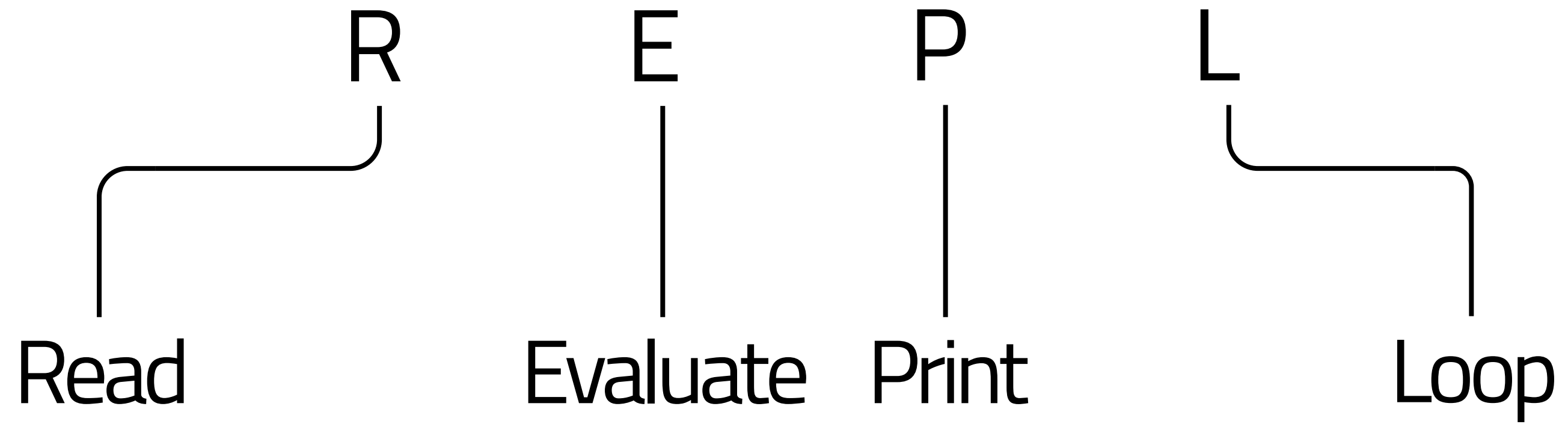
1 . x . x

الفردي - تحت التطوير

NVM

<https://github.com/nvm-sh/nvm>

```
nvm install --lts
```



مدير الحزم NPM



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package.json

package-lock.json

dev dependencies

Scripts

1.0.0

Major Version

Breaking change,
e.g. a rebrand,
feature set
added

Minor Version

Non-breaking
noteworthy change,
e.g. new component,
updated styles

Patch Version

Small request or
bug fix, e.g. update
or edit existing
elements

Symbol	
*	Match any version
^ version	only accept minor releases
~ version	only accept patch releases
>, >=, <=, <	accept versions greater/less/equal to version.
1.2.x	1.2.0, 1.2.1 but not 1.3.0

الوحدات MODULES

التصدير الافتراضي

مسموح فقط بواحد تصدير افتراضي

تقدر تستورد بالاسم اللي انت عايزه

export default

التصدير بالاسم



Bun